
MEMS

An Open-Source Summary

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Used notebook LM on previous exam questions.

0.0.1 Part 1: Pressure Sensors

Can you use lateral plate motion for capacitive pressure sensors? How do you fabricate this?

No, this is generally not done. Pressure acts normal to the surface, causing a membrane to deflect vertically (gap change) rather than laterally. Lateral motion (changing overlap area) is typically used for accelerometers (comb drives).

- *Note:* While you *could* design complex linkages to convert vertical deflection to lateral motion, it is inefficient compared to the standard parallel plate variable-gap method.

Can you use differential sensing for capacitive pressure sensors?

It is difficult to implement a *true* mechanical differential structure (where one gap increases and another decreases simultaneously) because the pressure usually pushes the membrane in only one direction (inwards). If you sandwich the two wafer, one will have a hole to let the pressure in which will fundamentally changes the capacitor value.

- *Contrast:* Accelerometers move bidirectionally, making true differential sensing easy.

How can you create an “almost” differential capacitive pressure sensor?

You typically use a **reference capacitor** placed on the same die. This reference capacitor is designed to be insensitive to pressure (rigid) but experiences the same temperature and environmental conditions as the sensing capacitor.

- *Readout:* You measure the difference between the active sensing capacitor (C_{sense}) and the reference capacitor (C_{ref}) to cancel out common-mode errors.

Can you use piezoelectrics for pressure sensors?

Yes, but **only for dynamic pressure** (e.g., blast waves, microphones).

- *Limitation:* Piezoelectric materials generate charge only when the stress *changes*. The charge eventually leaks away, making them incapable of measuring static (DC) ambient pressure.

What’s the problem with using piezoelectric pressure sensors for ambient pressure?

They have **no DC response**. Because the generated charge leaks off over time (determined by the RC time constant), the signal decays to zero even if the pressure remains constant.

0.0.2 Part 2: Accelerometers & Dynamics

What physical parameters influence the sensitivity, bandwidth, and dynamic range?

- **Sensitivity (S_0):** Scales with $\frac{m}{k}$. Larger mass (m) or softer spring (k) increases sensitivity.
- **Bandwidth (BW):** Approximately equal to the resonant frequency (ω_n), which scales with $\sqrt{\frac{k}{m}}$. Stiffer springs or lighter masses increase bandwidth.
- **Dynamic Range:** Limited at the low end by **Brownian noise** ($\text{Noise} \propto \sqrt{\frac{b}{m}}$) and at the high end by linearity limits or mechanical stops.

What is the downside of having large m and small k ?

While sensitivity increases, the **Bandwidth decreases** significantly (ω_n drops). Additionally, the proof mass may sag excessively under gravity, potentially causing nonlinearity or hitting the mechanical stops.

How to solve the problem of large displacements? Which approach is used?

Use **Closed-Loop Operation (Force Feedback)**.

- *How it works:* An electronic feedback loop applies a voltage to the electrodes to generate an electrostatic force (F_f) that opposes the inertial force (ma). This keeps the mass at the “null” (center) position.

How can I increase the Dynamic Range and prevent the proof mass from hitting electrodes?

Implement the **Closed-Loop System** described above. Because the mass barely moves (it is “force-balanced”), you avoid the non-linear regions of the capacitor gap and prevent physical contact with the electrodes.

How can I implement this system?

You need a servo loop containing:

1. **Sensing Element:** Detects small displacement (x).
2. **Amplifier/Integrator:** Converts ΔC to Voltage.
3. **Controller (PID or $\Sigma\Delta$):** Determines the necessary feedback force.
4. **Actuator:** Applies feedback voltage to the electrodes to re-center the mass.

0.0.3 Part 3: Fabrication & Design

If you had a MEMS lab and an SOI wafer, how would you fabricate an accelerometer?

I would use a **Bulk Micromachining** process using **DRIE (Deep Reactive Ion Etching)**.

1. **Material:** SOI wafer (Handle = Substrate, Oxide = Sacrificial, Device = Structural mass).
2. **Patterning:** Use lithography and DRIE on the Device layer to define the proof mass, springs, and comb fingers with vertical sidewalls.
3. **Release:** Use HF (Hydrofluoric acid) to etch the sacrificial oxide layer underneath the moving parts, freeing the mass.

What are typical thicknesses of proof mass?

- **Surface Micromachining:** Thin, typically **2–4 μm** (deposited polysilicon).
- **Bulk Micromachining:** Thick, typically full wafer thickness or **300–500 μm** (single crystal silicon).

What's the downside of the slopes of KOH etching? What can you use instead? What kind of etch stop?

- **Downside:** KOH is anisotropic, creating **54.7° slopes** which consume valuable chip area (large footprint for small bottom features).
- **Instead:** Use **DRIE** (Deep Reactive Ion Etching) for vertical (90°) sidewalls and high aspect ratios.
- **Etch Stop (for KOH):** Use a **Boron ($p++$) etch stop** or an **Electrochemical Etch Stop (ECE)** at a p-n junction (creates a N profile at cathode).

0.0.4 Part 4: Comparisons & Selection

Compare capacitive and piezoresistive sensing.

Feature	Capacitive	Piezoresistive
Power	Low power	High power (current must flow)
Temp Stability	Good (cancels in differential)	Poor (requires compensation)
Noise	High (if mass is small - Brownian)	Low/Moderate (Johnson noise)
Integration	Complex electronics required	Simple readout (Wheatstone bridge)
DC Response	Yes	Yes

[Source 6, 50]

Which one would you use in automotive and which in a human body?

- **Automotive (Airbag): Capacitive** (Surface micromachined, e.g., ADXL). It is cheap, reliable, and allows for monolithic integration of self-test electronics on the same chip.
- **Human Body (Blood Pressure): Piezoresistive** (e.g., Novasensor). It is low-cost, disposable, reliable in fluid environments, and provides a low-impedance output that is less susceptible to noise interference inside the body compared to capacitive.
 - mmmh yeah but high power consumption how do you want to recharge it ???
- For harsh environment where T is high and wide + EMC is important → Use a **piezoelectric** sensor.

0.0.5 Part 5: Design Scenario (Your specific “Sensor to make”)

What sensor do you wanna make using this technique (SOI)?

A **High-Precision In-Plane Accelerometer** (using Bulk Micromachining).

How do you make this sensor?

Using the SOI process described in Q8: DRIE etching of the device layer to form interdigitated comb fingers, followed by an HF release of the buried oxide.

What are the design parameters?

1. **Mass (m):** Maximized by using the thick SOI device layer (e.g., $50\ \mu\text{m}$).
2. **Spring Constant (k):** Tuned via beam width/length to set ω_n .
3. **Damping (b):** Controlled by the air gap between fingers (squeeze film damping) to achieve $Q \approx 0.7$.
4. **Capacitive Overlap (A):** Maximized by using many comb fingers.

If I make the mass bigger what will happen to the system?

- **Sensitivity:** Increases.
- **Noise:** Decreases (Brownian noise drops).

What happens with the bandwidth?

- **Bandwidth:** Decreases (Resonant frequency drops).

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